



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 02.08.2026

In addition to part -I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the course.

Course No.	: CE F320
Course Title	: DESIGN OF REINFORCED CONCRETE STRUCTURES
Instructor-in-charge	: MANOJ KUMAR
Tutorial Instructors	: Rajesh Kumar, Suresh S (RS), and Vignesh Ganesan (RS)

1. Course Description:

Design Philosophies: Introduction to working stress design philosophy; Limit State design philosophy, Limit State design of Singly reinforced and doubly reinforced rectangular and flanged section beams; Design of one-way and two-way slabs in flexure; Design of beams and slabs for shear; Design of beams and slabs for bond, anchorage and development length; Design of beams and slabs for serviceability; Limit state design of columns subjected to axial load, and axial load plus uni-axial and bi-axial bending; Design of footings under the columns; Design of simple staircases.

2. Scope and Objective of the Course

In this course, the students will be introduced to the various design philosophies used in the design of Reinforced Concrete (RC) structures, with a particular focus on the Limit State Design method. The course will provide an in-depth discussion of the concepts of Limit State Design method. Subsequently, the Limit State Design method will be employed to the design of singly and doubly reinforced rectangular- and T-section beams and one-way and two-way slabs, for flexure, shear, anchorage, bond, and serviceability requirement. Additionally, the course will cover the design of columns subjected to axial loads, and axial load plus uniaxial and biaxial moments. Moreover, the course will explore different types of staircases, with a specific emphasis on the design of dog-legged staircases. At the end of this course, students will be able to design the basic structural elements of an RC buildings.

3. Text Book and References

TB: S. Unnikrishan Pillai and Devdas Menon, “Reinforced Concrete Design”, 4th Ed., 2022, McGraw Hill, New Delhi

RB1: Subramanian, N., “Design of Reinforced Concrete Structures,” Oxford Univ. Press, 2013, New Delhi

RB2: IS 456:2000 “Code of practice for Plain and Reinforced concrete”, BIS, New Delhi

RB3: Special Publication SP-16: Design aids for Reinforced Concrete to IS 456:1978, BIS, New Delhi.



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4. Course Plan:

Module	Lec. No.	Ref. to TB	Learning outcomes
Introduction	1-3	Ch. 1-2	Objectives of Structural Design, structural systems, RC buildings, Structural analysis and design, Design codes and Handbooks, characteristic Load, Constituent Materials of Reinforced concrete, properties of concrete and steel including stress-strain curve
Concepts and Principles of Limit state Design	4-5	Ch. 3	Design Considerations, introduction to different design philosophies, concept of Limit state design method, design stress-strain curves for concrete and steel
Analysis of beams and one-way slab for Flexure	6-10	Ch.4: 4.7-8	Analysis of singly and doubly reinforced rectangular and Flanged section beams and slabs for flexure at Ultimate Limit State
Design of beams and one-way slab for Flexure	11-16	Ch. 5	Requirements for Flexural Reinforcement and Deflection control, guidelines for selecting the section dimensions, Design of singly reinforced Rectangular section beams and one-way slabs, Design of continuous one-way slabs, Design of doubly reinforced Rectangular section beams, Design of singly and doubly reinforced Flanged section beams, curtailment flexural tension reinforcement.
Design of Flexural members for Shear	17-20	Ch. 6	Shear stresses in beams, behaviour of reinforced concrete beam under shear, Critical Section for Shear Design, Design shear strength of plain concrete and reinforced concrete, Additional comments on shear Reinforcement Design,
Design of Flexural Members for Bond	21-24	Ch. 8	Mechanism of bond and types of bond: Flexural and Anchorage bond, design of flexural members for bond
Design of Flexural members for Limit State of Serviceability	25-28	Ch. 10	Serviceability Limit State of deflection, short term deflection, long term deflection, Serviceability Limit State of flexural cracking,
Design of two-way Slabs	29-33	Ch. 11: 11.1-3	Two-way slab systems, Design of two-way slabs supported on walls, Design of two-way slabs supported on beams,
Design of Staircases	34-36	Ch. 12	Types of staircases, structural Classification of staircases, Loads on stair slabs, Design of Transversally spanning staircases, Design of Longitudinally spanning staircases,
Design of Compression Members	37-42	Ch. 13: 13.1-5	Classification of columns, estimation of effective length of columns, code requirements for the columns, Design of axially loaded short columns, Design of short columns under axial load plus uniaxial bending.

5. Evaluation Scheme

S. N.	Evaluation Component	Weightage	Duration	Day & Hr./ Date & Time	Comments
2.	Evaluative Tutorials	25%	50 Min	W1	Open Book [TBA]
3.	Mid-semester Exam.	30%	90 Min	8/10 FN1	Close Book
4.	Comprehensive Exam.	45%	180 Min	1/12 FN	Close Book

6. Chamber Consultation Hour: To be announced in the class.

7. Notice: Notice if any, concerning this course will be communicated via Nalanda or via institute e-mail.

8. Make-up Policy: Make-up will be granted only to genuine cases. For cases related to illness, prior permission with proper documentary evidence is essential.

Instructor-in-charge, CE F320